
CMP6200

Individual Undergraduate Project

Project Final Progress Review

**BCG: A Game-Based Approach to Motor
Imagery Adaptation and BCI Calibration**

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1.0 Progress to Date - Summary

The project is now in the model training and game design phase, having successfully completed all foundational infrastructure work since the Interim Progress Review in January 2026. First, four EEG devices were investigated and compared, leading to the selection of the Emotiv EPOC X as the primary experimental platform. The device is now fully configured, with a Python pipeline successfully streaming real-time raw EEG data via the Cortex API. Second, I used the MIMED public dataset to train two classification pipelines (CSP+LDA and CSP+SVM). Third, two game concepts were initially defined for the gamified calibration interface. And finally, our abstract on the BCG approach was accepted at the BME 11 conference. The most significant finding to date is that cross-subject classification accuracy on MIMED ranges from 42% to 52%, below the threshold required for reliable BCI control. While MIMED is the most hardware-compatible dataset available, having been collected on the same Emotiv EPOC X device, its limited trial pool of 240 samples is insufficient for CSP to generalise across subjects. This has prompted a strategic review of the training approach, with larger benchmark datasets and pretrained deep learning models currently being implemented.

2.0 Artefact Design and Development

2.1 Implementation and Current State

The BCG system is built around a core insight: if a game's objectives naturally require motor imagination, every in-game action becomes a labelled calibration trial without explicit prompting. The system has three components.

EEG Data Acquisition. The Emotiv EPOC X was selected following a hands-on comparison of four consumer EEG devices. Its 14-channel layout reliably covers the C3/C4 motor cortex positions critical for Motor Imagery classification. Following a direct discussion with the Emotiv support team, Emotiv sponsored a six-month research licence granting access to raw EEG data via the Cortex API. A custom Python pipeline now streams this data in real time.

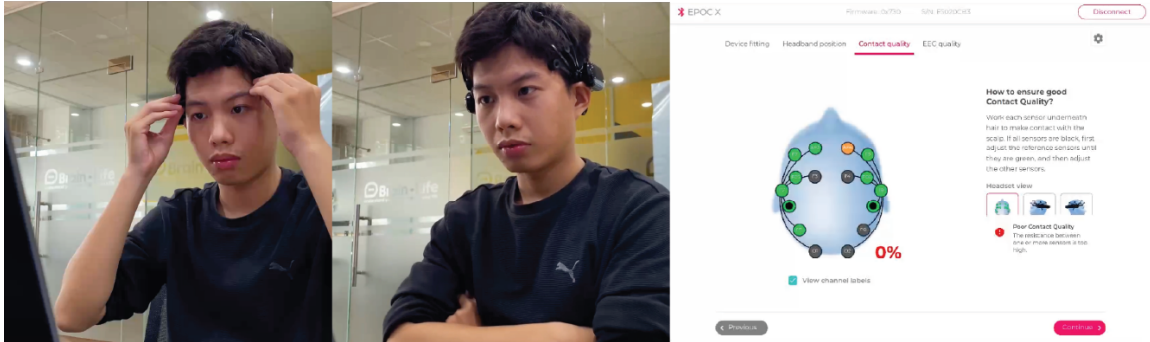


Figure 1: Author wearing the Emotiv EPOC X during device testing

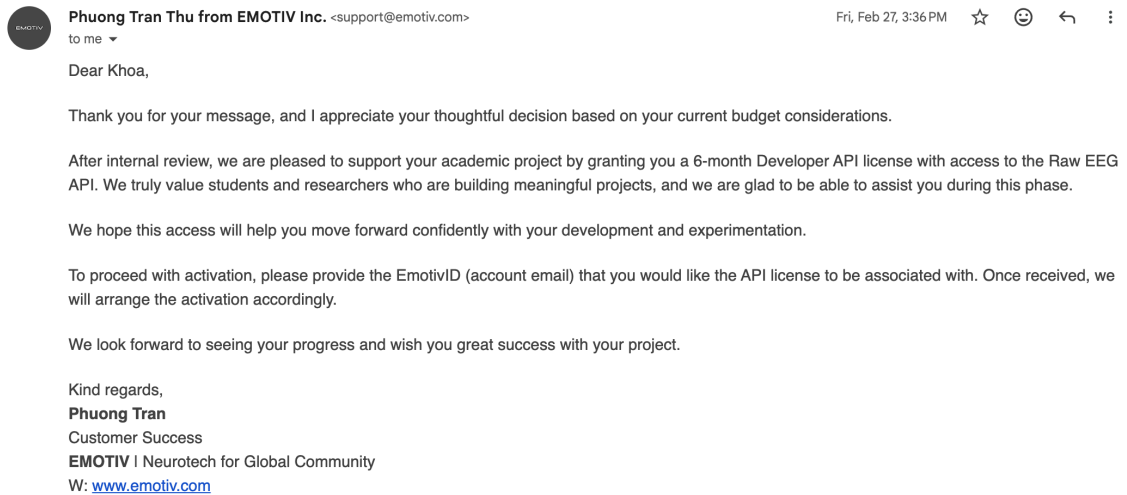


Figure 2: Emotiv Developer API license approval email

Signal Processing and Classification. The initial pipeline bandpass filters raw EEG to 8-30 Hz, applies CSP for feature extraction, and classifies using LDA or SVM. MIMED was selected as the initial training dataset as it was recorded on the same Emotiv EPOC X device, minimising the domain gap. Benchmarked results are shown in Table 1.

Model	Dataset	Evaluation	Accuracy
CSP + LDA	MIMED	Cross-subject, 5-fold	52.5%
CSP + SVM	MIMED	Cross-subject, 5-fold	46.7%
CSP + LDA	MIMED	Within-subject	42.1%

Table 1: Initial classification results on the MIMED dataset

The results, however, fall below the 70% BCI control threshold due to MIMED’s limited scale of 240 trials. Retraining on PhysioNet EEGMMIDB or fine-tuning EEGNetv4 are currently being evaluated.

Game Interface. The game interface is being developed in GameMaker Studio 2, expected to be connected to the Python pipeline via a custom Cortex API bridge. The game is designed to make gameplay naturally drive motor imagination, with labelled calibration data collected implicitly in the background.

2.2 Analysis and Reflection

The key realisation from this phase is that BCG’s value lies not in building a better classifier, but in investigating whether natural, game-driven motor imagination can replace the repetitive cue-based calibration protocol. This reframes the design priority: the game must feel spontaneous enough that the player’s MI signals are clean and consistent. Forced or distracted imagination produces weaker ERD/ERS patterns; motivated, relaxed gameplay produces better data. This is why the game design is deliberately simple, complexity would compete for the same cognitive bandwidth the BCI system needs to read.

The MIMED results (42-52%) reflect a cross-subject baseline on a small dataset, not the final performance target. In the full system, the classifier will be updated continuously using labelled data collected during gameplay, adapting to each user over time. The low initial accuracy is expected, and the implicit collection mechanism is precisely how it will be addressed

3.0 Updated Planning for Project Completion

3.1 High-priority Tasks

The following high-priority tasks remain to complete the project:

- Finalise the classification pipeline by evaluating PhysioNet EEGMMIDB and EEG-Netv4 as training alternatives
- Complete the BCG game prototype in GameMaker Studio 2 with left/right MI-driven controls

- Build and test the real-time Python-GameMaker communication bridge via the Cortex API
- Design the traditional baseline calibration protocol for the comparative user study
- Conduct a within-subjects user study with 8-10 participants, measuring classification accuracy and NASA-TLX workload
- Write and finalise the full dissertation
- Prepare the project poster and viva demonstration

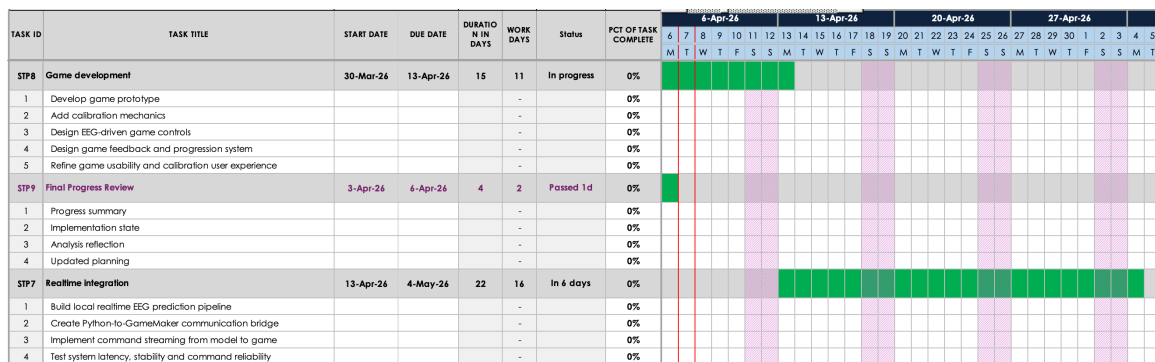


Figure 3: Project Timeline - April

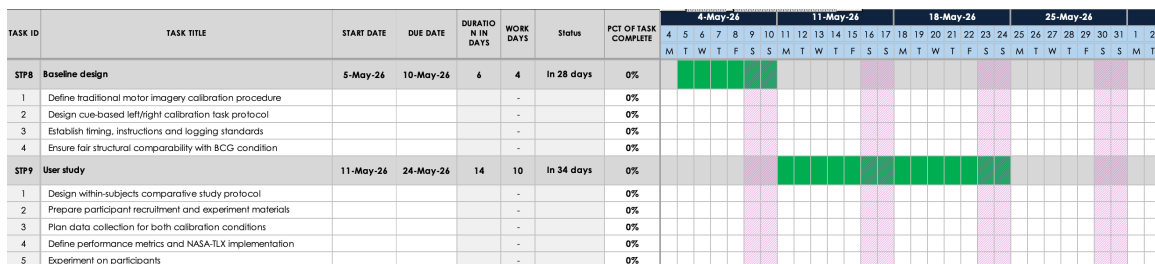


Figure 4: Project Timeline - May

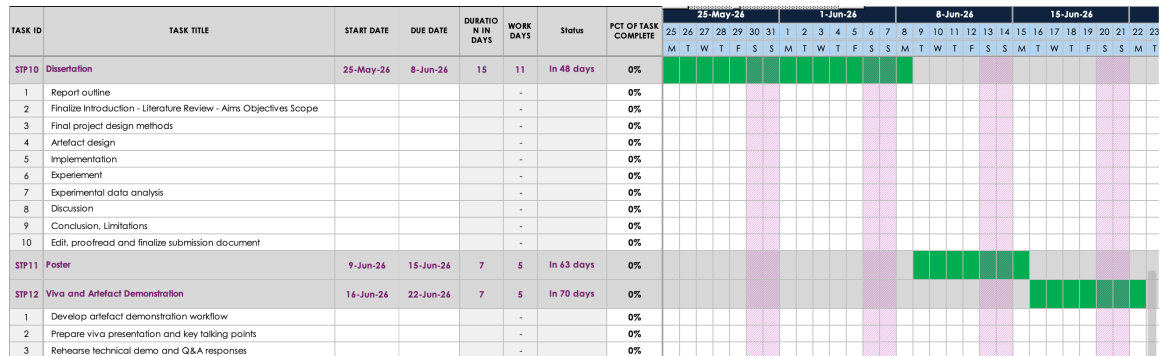


Figure 5: Project Timeline - June

3.2 Draft Outline of Final Report

1. Introduction
2. Review of Existing Knowledge
3. System Requirements and Architecture
4. EEG Hardware Investigation and Device Selection
5. Signal Processing and Classification Pipeline
6. BCG Game Design and Implementation
7. Real-Time System Integration
8. Baseline Protocol Design
9. User Study and Evaluation
10. Discussion
11. Conclusion and Future Work